

AI Agents for Cognitive Augmentation

Instructors	Valdemar Danry and Professor Pattie Maes
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Format	Weekly 2-hour session: 1-hour lecture + 1-hour hands-on tutorial
Course structure	Classes, weekly technical implementation assignments, cumulative agentic website, final project or study
Credits	2-0-10
Time	11am-1pm
Classroom	E15-341
First class	9/15, T

Course Overview

AI Agents for Cognitive Augmentation is a hands-on graduate-level course exploring the design, implementation, evaluation, and human impact of AI agents. AI systems are shifting from passive tools that respond to prompts toward agentic systems that take actions in the world by perceiving the environment, planning, use of tools and coordination with people. The course will consider how agents can be used to augment human cognitive capabilities.

Students will learn to build agents using state-of-the-art large language models and frameworks, connect agents to multimodal inputs, design test environments, fine-tune and improve agent behavior, and critically evaluate how agents can integrate with or disrupt human cognitive processes. The course will treat agents not as a single technology, but as a broad design space: agents that play video games, operate machines or robots, manage calendars, browse the web, control software, coordinate organizations, build code, improve websites, run simulated businesses, or assist humans in thinking, learning, and decision-making.

The course will ask a central question: **how can agents augment human cognitive abilities rather than replacing or weakening them?** Alongside technical agent-building, students will study how AI agents affect attention, memory, learning, reasoning, creativity, agency, and social behavior. A dedicated module will focus on BCIs, wearables, and physiological sensing, giving students hands-on experience building agents that respond to human cognitive and bodily signals.

Motivation

AI agents are becoming one of the defining applications of artificial intelligence. Yet the field is still young: current agents are powerful but brittle, impressive but unreliable, increasingly autonomous but poorly understood. This creates a rare educational opportunity. Rather than only teaching established techniques, the course will function as an experimental studio and technical lab for discovering what agents can do today, where they fail, and how they might be designed more responsibly.

The Media Lab is uniquely positioned to host this course. The Lab has a long history of work on software agents, human augmentation, wearable computing, and human-centered AI. Co-taught with Pattie Maes, whose work helped define the field of software agents, the course will connect this history to the current frontier of LLM-based, multimodal, embodied, and socially embedded agents.

Learning Goals

By the end of the course, students will complete some of the following based on their interests:

1. Build functional AI agents using large-language models, tool-calling, memory, planning, and environment loops.
 2. Connect agents to multimodal inputs such as vision, audio, wearables, or interaction traces.
 3. Design and evaluate agent environments, benchmarks, simulations, and failure cases.
 4. Understand techniques for improving agent performance, including fine-tuning, reinforcement learning approaches such as GRPO, preference optimization, and iterative evaluation.
 5. Prototype agents within a chosen domain that could include games, robotics, software control, organizations, personal productivity, education, and cognitive augmentation.
 6. Critically analyze the effects of AI agents on human cognition, agency, learning, overreliance, and social behavior.
 7. Design agents that augment human capabilities rather than simply automate human tasks.
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Course Structure

The course will meet once per week for two hours. Each session will include a one hour lecture covering concepts, systems, papers, and design principles, followed by a one hour hands-on tutorial where students implement technical components.

Every week, students will complete an open-ended agent build based on the lectures and tutorials. Similar in spirit to *How to Make (Almost) Anything*, the assignments will emphasize technical demonstration, creative interpretation, and personal exploration. Students will maintain an “agentic website” and GitHub repository throughout the semester, updating them with each assignment and documenting what they built, how it works, what failed, and what they learned. They will show each build and present its technical implementation in class.

The semester will culminate in a final project: either a substantial agentic system or an empirical study evaluating an agent's capabilities, limitations, or effects on people.

The course will enroll 30 for-credit students and up to 10 listeners. Listeners participate in course sessions and weekly builds on the same basis as for-credit students but are not graded. Priority will be given to students from Media, Arts and Sciences and other MIT programs over students from other institutions.

Modules

Possible course modules include:

- Foundations of agents: perception, planning, action, memory, tools, environments
- Running and fine-tuning large-language models (for agentic tasks)
- Building API-based agents and tool-using systems
- Computer-using agents and agents that operate software

- Browser, coding, and website-improvement agents
- Agents in video games and simulated worlds
- Generative agents and multi-agent societies
- Agent evaluation, benchmarks, sandboxes, and adversarial test environments
- Multimodal agents with vision, audio, and interaction data
- Reinforcement learning for agent improvement, including GRPO-style methods
- Wearables, BCIs, and physiology-aware agents
- Cognitive augmentation: agents for reasoning, learning, creativity, reflection, and decision-making
- Human risks: overreliance, deskilling, persuasion, privacy, autonomy, and delegation

Assignments and Evaluation

Students will be evaluated through a sequence of open-ended, weekly technical implementations, documentation on their agentic website, in-class demos, peer critique, and a final project or study. The emphasis will be on working systems, thoughtful experimentation, clear documentation, and honest analysis of limitations.

Each assignment will ask students to demonstrate a specific technical skill while leaving the application domain open. For example, after learning tool-calling and memory, one student might build a calendar agent, another might build a game-playing agent, and another might build a research assistant that updates a website. After the multimodal module, one student might build a gaze-aware reading assistant while another builds an agent that responds to sound, movement, or visual context.

For the final projects, students will work in small groups and build a functioning, novel agent application or technique. They will also document the why, what, how of their project. The final project will be evaluated based on both its novelty and strength of the idea as well as on its execution, completion and writeup.

Component	Weight
Weekly technical builds	35%
Agentic website / documentation	15%
In-class demos and critique	15%
Final project system or study	25%
Final paper-style report	10%